DARPA-PA-26-05 MATHBAC

Abstract Summary Slide Template

SCBE Mathematical Framework for Agentic Communication Protocols

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CONCEPT

Composed operator $T = L_{14} \circ \cdots \circ L_1$ acting on:

$\psi(t) = (\psi_1(t),\dots,\psi\_N(t), G\_t, pd(t)) \in B^{n \times N} \times G\_N \times \mathbb{R}_+$

Agent state $\psi\_i(t)$ embedded in Poincaré ball B^n via L4 (exponential map).

Trust-weighted directed Laplacian: $L\_H(G\_t)$, edge $w\_ij = H(d\_H(\psi\_i,\psi\_j), pd\_ij)$

Harmonic wall (L12): $H(d,pd) = 1/(1 + \varphi \cdot d\_H + 2 \cdot pd) \in (0,1]$

CDPTI = $\mathrm{Re}(\lambda_2(L\_H(G\_t))) / \mathrm{Re}(\lambda_2(L\_H(G\_t0)))$ — signed coherence drift

Safety is a geodesic property of T, not a post-hoc filter.

BAA alignment: TA1, MC-1 through MC-5

APPROACH

PI: Issac Davis — SCBE-AETHERMOORE (sole prop, SAM-active; UEI J4NXHM6N5F59)

Sub: Hoags Inc. (bounded DAVA corroborating support, if finalized)

Benchmark evidence:

- Petri adversarial: 172/173 seeds (99.42%); 0.58% false-allow after regex pre-filter
- Semantic sphere: 26/26 surfaces, 505 iterations (2026-05-30)
- Terminal-bench-core-0.1.1: 13/13 (zero governance overhead)
- Real-patch task harness: 5/5 SCBE vs. 0/5 baseline

Independent derivation: Lyapunov stability, CBF, port-Hamiltonian, persistent excitation, gyroscopic precession (Khalil; Marsden/Ratiu; Cannon)

Prior government contracts: None | SAM: Active

RISKS

Phase I Objectives:

1. Operator formalization — Lyapunov constants η, b; NMR task families (M3)
2. CDPTI ≥ 95% recall, ≤ 5% FPR vs. Mixtral-8x7B baseline (M13)
3. Harmonic wall cost curve — φ-weighting ablation; adversarial overhead ratio (M9)
4. NMR validation — Task A (1H NMR→structure) + Task B (2D COSY); ChemBERTa-77M + Qwen2.5-Coder-0.5B; Hammett lane (M9)
5. Four PA metrics — MEE, ACV, CDPTI, PIS; IV&V-computable from JSONL bundle

Why composed operator:

Spera (2026, arXiv:2603.15973): 42.6% of 900 real trajectories have conjunctive multi-agent dependencies — T addresses directly via geometric accumulation.

Flat scores, behavioral contracts, post-hoc red-teaming all fail at protocol-graph level.

TEAM

Phase I Schedule (16 months, OT §4021):

M1 (mo. 1) — Baselines; SSM calibration; IV&V bundle format

M3 (mo. 3) — Lyapunov constants; L_H(G_t) instrumented; CDPTI live

M6 (mo. 6) — CDPTI demo; ≥2 NMR variants; PI meeting

M9 (mo. 9) — Software suite; Hammett generalization; ROMs

M13 (mo.13) — Baseline vs. Mixtral; PIS catalog v1; PI meeting

M14 (mo.14) — Computational design tool (CDT)

M16 (mo.16) — Final report + Phase II plan

Key metrics:

MEE: emission rate (calibrated M3) | ACV: mean ≥0.90 at M9

CDPTI: ≥95% recall | PIS: cluster separation at M13

Cost: $839,000 (42% of $2M cap) | Phase II ROM: ~$382,000

CAGE: 1EXD5 | Award: OT for Research (10 U.S.C. §4021)